George Pu | Resume

904-316-1566 pu.george99@gmail.com georgerpu.github.io george-r-pu GeorgeRPu

Applied Scientist at Amazon building AI agents and ML systems at scale. Published researcher in deep learning and federated learning. 4+ years of experience spanning research, ML infrastructure, and software development.

Experience

Amazon, Sustainability Science and Innovation **Seattle**
Applied Scientist *Jul 2025 – present*

- Developed deep research agent that auto-generates bills of materials, extending actionable environmental impact analysis to 75% of Amazon's carbon footprint with a <30% MAPE.
- Led sustainability scientists to annotate hundreds of life cycle inventories/assessments, creating 2 new internal benchmarks.

Amazon, Customer Trust **Seattle**
Machine Learning Engineer / Applied Scientist *Oct 2024 – Jul 2025*

- Created time series classification models that helped reduce bad seller debt by an estimated $3 million
- Trained LightGBM models to reduce investigator case load, saving an estimated $500,000 per year.
- Trained graph machine learning models on SageMaker.

Software Developer/Engineer *Jul 2021 – Oct 2024*

- Automated bad actor investigation case creation, saving 2 hours of investigator time per week.
- Onboarded ML models to SageMaker, saving >184 hours annually of scientist effort retraining models and batch inferencing.
- Designed and implemented an offline graph query execution engine over hundreds of millions of nodes on Amazon Neptune that helped stop $500,000+ in fraudulent sales; work featured on the AWS Database Blog

University of Florida **Gainesville**
Research Assistant *Sep 2017 – May 2021*

- Collaborated with a PhD student and professor to fit multi-compartment epidemiological models to Leishmaniasis data.
- Published 5 papers at top venues (IROS, NeurIPS) with over 250+ citations.

Selected Publications

Seeing Through Walls: Real-Time Digital Twin Modeling of Indoor Spaces: George Pu, Paul Wei, Amanda Aribe, James Boultinghouse, Nhi Dinh, Fang Xu, Jing Du. *2021 Winter Simulation Conference*.

Communication-Efficient Federated Learning via Dataset Distillation: Yanlin Zhou*****, George Pu*****, Xiyao Ma, Xiaolin Li, Dapeng Wu. *2020 NeurIPS Workshop on Scalability, Privacy, and Security in Federated Learning (SpicyFL)*.

Adaptive Leader-Follower Formation Control and Obstacle Avoidance via Deep Reinforcement Learning: Yanlin Zhou*****, Fan Lu*****, George Pu*****, Xiyao Ma, Runhan Sun, Hsi-Yuan Chen, Xiaolin Li. *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

Technical Skills

Languages: Java, Javascript, Python, SQL, Typescript

Python Packages: AutoGluon, Langchain, Matplotlib, Numpy, Pandas, Polars, PyTorch, Scikit-learn, Tensorflow

AI/ML: Agents, LLM Evaluation, Deep Learning, Graph Neural Networks, Prompt Engineering, RAG, Tabular Machine Learning, Time Series Classification

Cloud: AWS, Bedrock, CloudWatch, DynamoDB, IAM, Neptune, S3, SageMaker, Step Functions

Education

University of Florida **Gainesville**
B.S. Computer Science, 3.8/4.0 *Aug 2017 – May 2021*